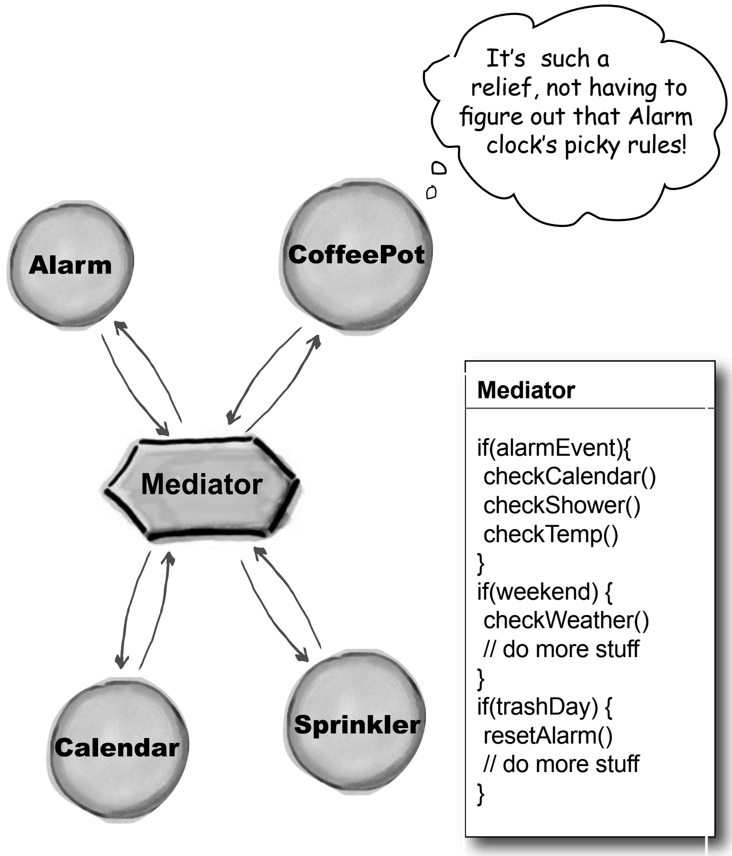
Mediator in action...

With a Mediator added to the system, all of the appliance objects can be greatly simplified:

- They tell the Mediator when their state changes.
- They respond to requests from the Mediator.

Before adding the Mediator, all of the appliance objects needed to know about each other... they were all tightly coupled. With the Mediator in place, the appliance objects are all *completely decoupled* from each other.

The Mediator contains all of the control logic for the entire system. When an existing appliance needs a new rule, or a new appliance is added to the system, you'll know that all of the necessary logic will be added to the Mediator.



Mediator Benefits

- Increases the reusability of the objects supported by the Mediator by decoupling them from the system.
- Simplifies maintenance of the system by centralizing control logic.
- Simplifies and reduces the variety of messages sent between objects in the system.

Mediator Uses and Drawbacks

- The Mediator is commonly used to coordinate related GUI components.
- A drawback of the Mediator pattern is that without proper design, the Mediator object itself can become overly complex.

Memento

Use the Memento Pattern when you need to be able to return an object to one of its previous states; for instance, if your user requests an "undo."

A scenario

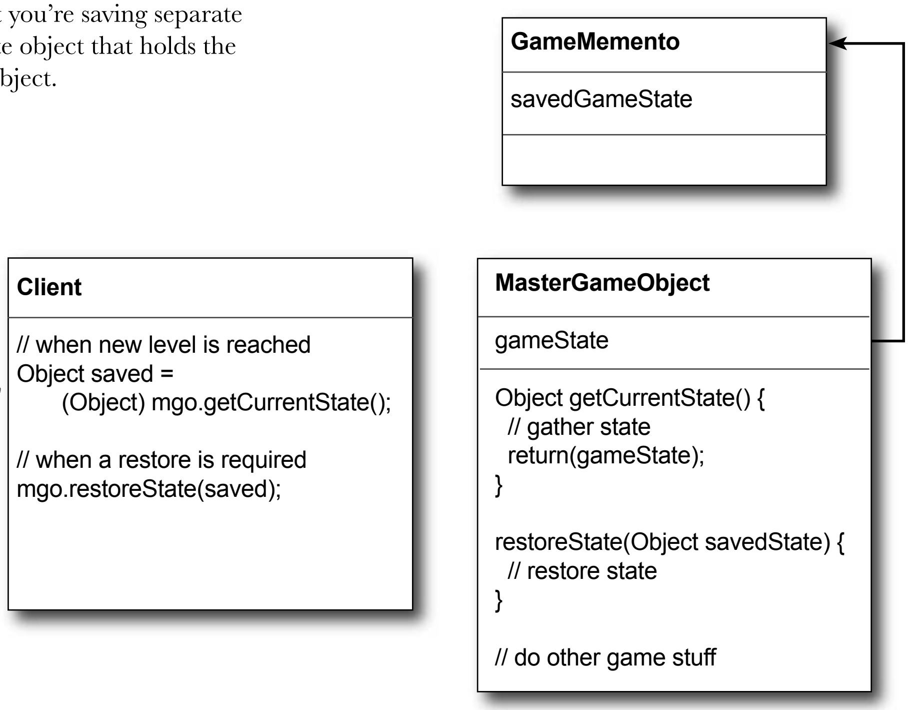
Your interactive role playing game is hugely successful, and has created a legion of addicts, all trying to get to the fabled "level 13." As users progress to more challenging game levels, the odds of encountering a game-ending situation increase. Fans who have spent days progressing to an advanced level are understandably miffed when their character gets snuffed, and they have to start all over. The cry goes out for a "save progress" command, so that players can store their game progress and at least recover most of their efforts when their character is unfairly extinguished. The "save progress" function needs to be designed to return a resurrected player to the last level she completed successfully.

The Memento at work

The Memento has two goals:

- Saving the important state of a system's key object.
- Maintaining the key object's encapsulation.

Keeping the single responsibility principle in mind, it's also a good idea to keep the state that you're saving separate from the key object. This separate object that holds the state is known as the Memento object.



While this isn't a terribly fancy implementation, notice that the Client has no access to the Memento's data.

Memento Benefits

- Keeping the saved state external from the key object helps to maintain cohesion.
- Keeps the key object's data encapsulated.
- Provides easy-to-implement recovery capability.

Memento Uses and Drawbacks

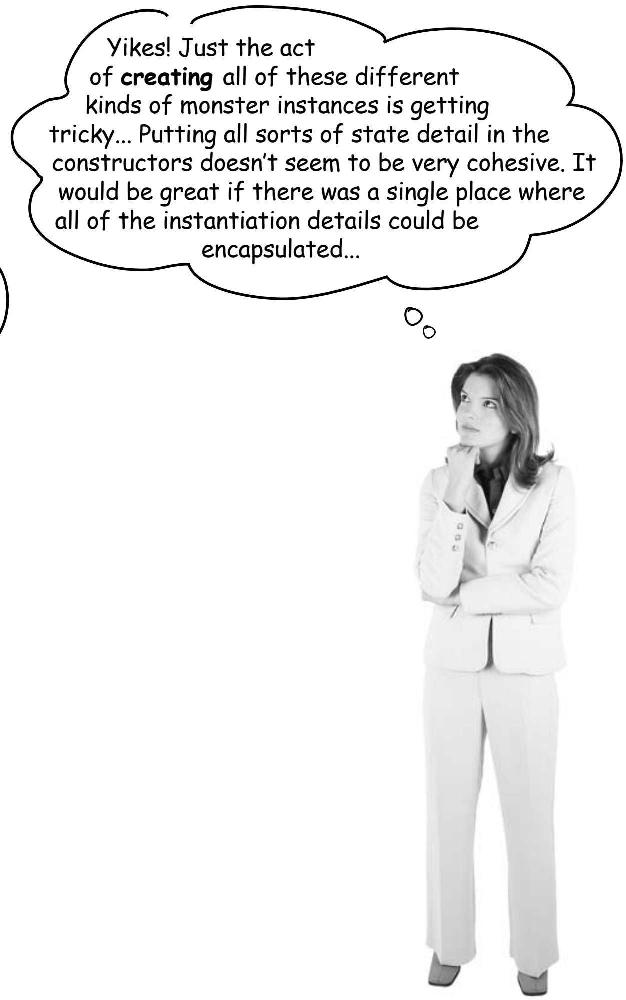
- The Memento is used to save state.
- A drawback to using Memento is that saving and restoring state can be time consuming.
- In Java systems, consider using Serialization to save a system's state.

Prototype

Use the Prototype Pattern when creating an instance of a given class is either expensive or complicated.

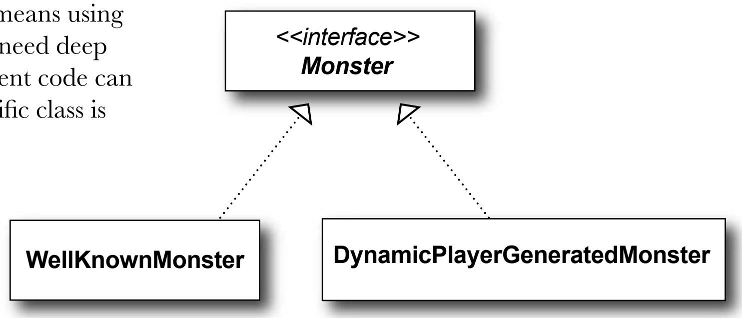
A scenario

Your interactive role playing game has an insatiable appetite for monsters. As your heros make their journey through a dynamically created landscape, they encounter an endless chain of foes that must be subdued. You'd like the monster's characteristics to evolve with the changing landscape. It doesn't make a lot of sense for bird-like monsters to follow your characters into underseas realms. Finally, you'd like to allow advanced players to create their own custom monsters.



Prototype to the rescue

The Prototype Pattern allows you to make new instances by copying existing instances. (In Java this typically means using the clone() method, or de-serialization when you need deep copies.) A key aspect of this pattern is that the client code can make new instances without knowing which specific class is being instantiated.



MonsterMaker

```
makeRandomMonster() {
  Monster m =
        MonsterRegistry.getMonster();
}
```

The client needs a new monster appropriate to the current situation. (The client won't know what kind of monster he gets.)

MonsterRegistry

```
Monster getMonster() {
  // find the correct monster
  return correctMonster.clone();
}
```

The registry finds the appropriate monster, makes a clone of it, and returns the clone.

Prototype Benefits

- Hides the complexities of making new instances from the client.
- Provides the option for the client to generate objects whose type is not known.
- In some circumstances, copying an object can be more efficient than creating a new object.

Prototype Uses and Drawbacks

- Prototype should be considered when a system must create new objects of many types in a complex class hierarchy.
- A drawback to using the Prototype is that making a copy of an object can sometimes be complicated.

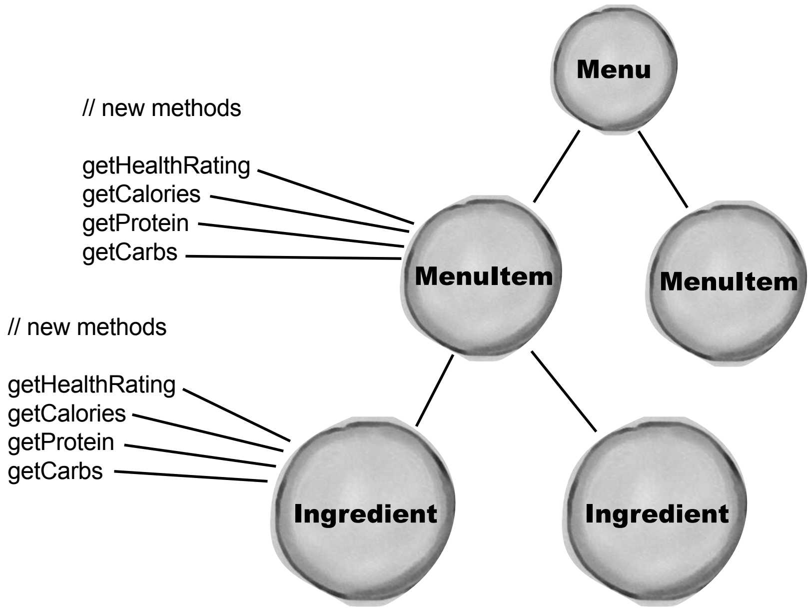
Visitor

Use the Visitor Pattern when you want to add capabilities to a composite of objects and encapsulation is not important.

A scenario

Customers who frequent the Objectville Diner and Objectville Pancake House have recently become more health conscious. They are asking for nutritional information before ordering their meals. Because both establishments are so willing to create special orders, some customers are even asking for nutritional information on a per ingredient basis.

Lou's proposed solution:

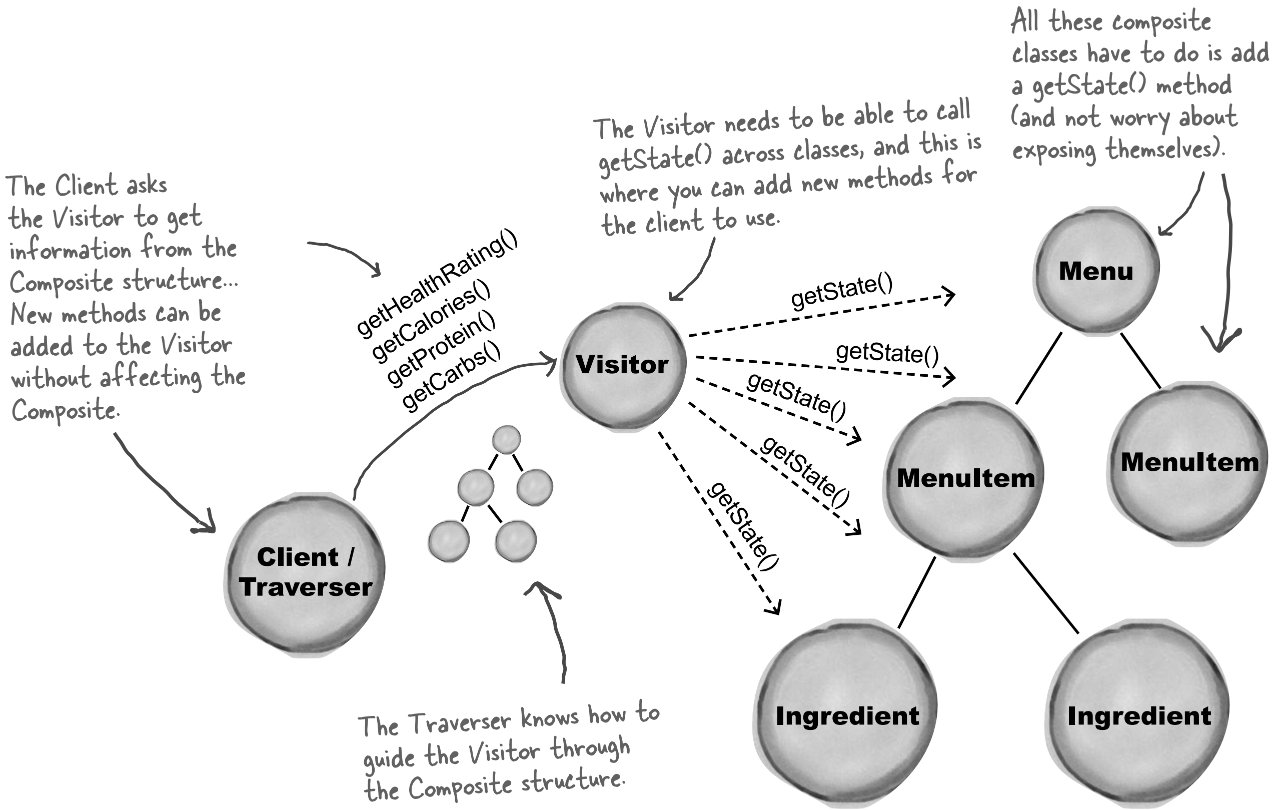


Mel's concerns...

"Boy, it seems like we're opening Pandora's box. Who knows what new method we're going to have to add next, and every time we add a new method we have to do it in two places. Plus, what if we want to enhance the base application with, say, a recipes class? Then we'll have to make these changes in three different places..."

The Visitor drops by

The Visitor must visit each element of the Composite; that functionality is in a Traverser object. The Visitor is guided by the Traverser to gather state from all of the objects in the Composite. Once state has been gathered, the Client can have the Visitor perform various operations on the state. When new functionality is required, only the Visitor must be enhanced.



Visitor Benefits

- Allows you to add operations to a Composite structure without changing the structure itself.
- Adding new operations is relatively easy.
- The code for operations performed by the Visitor is centralized.

Visitor Drawbacks

- The Composite classes' encapsulation is broken when the Visitor is used.
- Because the traversal function is involved, changes to the Composite structure are more difficult.

Index

D

N

O

P

Q

R

S

T

U

V

W

Y

Colophon

All interior layouts were designed by Eric Freeman, Elisabeth Freeman, Kathy Sierra and Bert Bates. Kathy and Bert created the look & feel of the Head First series. The book was produced using Adobe InDesign CS (an unbelievably cool design tool that we can't get enough of) and Adobe Photoshop CS. The book was typeset using Uncle Stinky, Mister Frisky (you think we're kidding), Ann Satellite, Baskerville, Comic Sans, Myriad Pro, Skippy Sharp, Savoye LET, Jokerman LET, Courier New and Woodrow typefaces.

Interior design and production all happened exclusively on Apple Macintoshes–at Head First we're all about "Think Different" (even if it isn't grammatical). All Java code was created using James Gosling's favorite IDE, *vi*, although we really should try Erich Gamma's Eclipse.

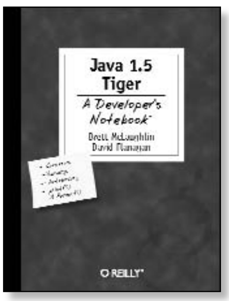
Long days of writing were powered by the caffeine fuel of Honest Tea and Tejava, the clean Santa Fe air, and the grooving sounds of Banco de Gaia, Cocteau Twins, Buddha Bar I-VI, Delerium, Enigma, Mike Oldfield, Olive, Orb, Orbital, LTJ Bukem, Massive Attack, Steve Roach, Sasha and Digweed, Thievery Corporation, Zero 7 and Neil Finn (in all his incarnations) along with a heck of a lot of acid trance and more 80s music that you'd care to know about.

And now, a final word from Head First Labs...

Our world class researchers are working day and night in a mad race to uncover the mysteries of Life, the Universe and Everything—before it's too late. Never before has a research team with such noble and daunting goals been assembled. Currently, we are focusing our collective energy and brain power on creating the ultimate learning machine. Once perfected, you and others will join us in our quest!

You're fortunate to be holding one of our first protoypes in your hands. But only through constant refinement can our goal be achieved. We ask you, a pioneer user of the technology, to send us periodic field reports of your progress, at fieldreports@headfirstlabs.com

Related Titles Available from O'Reilly



Java

Ant: The Definitive Guide
Better, Faster, Lighter Java
Eclipse
Eclipse Cookbook
Enterprise JavaBeans, *4th Edition*
Hardcore Java
Head First Java
Head First Servlets & JSP
Head First EJB
Hibernate: A Developer's Notebook
J2EE Design Patterns
Java 1.5 Tiger: A Developer's Notebook
Java & XML Data Binding
Java & XML
Java Cookbook, *2nd Edition*
Java Data Objects
Java Database Best Practices
Java Enterprise Best Practices
Java Enterprise in a Nutshell, *2nd Edition*
Java Examples in a Nutshell, *3rd Edition*
Java Extreme Programming Cookbook
Java in a Nutshell, *4th Edition*
Java Management Extensions
Java Message Service
Java Network Programming, *2nd Edition*
Java NIO
Java Performance Tuning, *2nd Edition*
Java RMI
Java Security, *2nd Edition*
JavaServer Faces
Java ServerPages, *2nd Edition*
Java Servlet & JSP Cookbook
Java Servlet Programming, *2nd Edition*
Java Swing, *2nd Edition*
Java Web Services in a Nutshell
Learning Java, *2nd Edition*
Mac OS X for Java Geeks
Programming Jakarta Struts *2nd Edition*
Tomcat: The Definitive Guide
WebLogic: The Definitive Guide

O'REILLY®

Our books are available at most retail and online bookstores.
To order direct: 1-800-998-9938 • *order@oreilly.com* • *www.oreilly.com*
Online editions of most O'Reilly titles are available by subscription at *safari.oreilly.com*

Keep in touch with O'Reilly

1. Download examples from our books

To find example files for a book, go to:

www.oreilly.com/catalog

select the book, and follow the "Examples" link.

2. Register your O'Reilly books

Register your book at *register.oreilly.com*

Why register your books?
Once you've registered your O'Reilly books you can:

- Win O'Reilly books, T-shirts or discount coupons in our monthly drawing.
- Get special offers available only to registered O'Reilly customers.
- Get catalogs announcing new books (US and UK only).
- Get email notification of new editions of the O'Reilly books you own.

3. Join our email lists

Sign up to get topic-specific email announcements of new books and conferences, special offers, and O'Reilly Network technology newsletters at:

elists.oreilly.com

It's easy to customize your free elists subscription so you'll get exactly the O'Reilly news you want.

4. Get the latest news, tips, and tools

www.oreilly.com

- "Top 100 Sites on the Web"—PC Magazine
- CIO Magazine's Web Business 50 Awards

Our web site contains a library of comprehensive product information (including book excerpts and tables of contents), downloadable software, background articles, interviews with technology leaders, links to relevant sites, book cover art, and more.

5. Work for O'Reilly

Check out our web site for current employment opportunities:

jobs.oreilly.com

6. Contact us

O'Reilly & Associates
1005 Gravenstein Hwy North
Sebastopol, CA 95472 USA

TEL: 707-827-7000 or 800-998-9938
(6am to 5pm PST)

FAX: 707-829-0104

order@oreilly.com
For answers to problems regarding your order or our products. To place a book order online, visit:

www.oreilly.com/order_new

catalog@oreilly.com
To request a copy of our latest catalog.

booktech@oreilly.com
For book content technical questions or corrections.

corporate@oreilly.com
For educational, library, government, and corporate sales.

proposals@oreilly.com
To submit new book proposals to our editors and product managers.

international@oreilly.com
For information about our international distributors or translation queries. For a list of our distributors outside of North America check out:

international.oreilly.com/distributors.html

adoption@oreilly.com
For information about academic use of O'Reilly books, visit:

academic.oreilly.com